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Mike Hall

Principal Software Engineer

Location Chicago, IL

Email mike@just3ws.com

Phone [\(847\) 877-3825](tel:(847)877-3825)

Web just3ws.com

LinkedIn linkedin.com/in/just3ws/

GitHub github.com/just3ws

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Technical Leadership & Production Systems

Modernization

Principal Software Engineer who modernizes high-consequence production systems. Helps engineering leaders make confident decisions about aging architectures by establishing clear domain ownership, OpenTelemetry distributed tracing, zero-downtime safety, and practical AI engineering across Ruby on Rails, PostgreSQL, and cloud infrastructure.

Areas of focus

01 Technical Leadership & Cartography

02 Legacy Systems & Modernization

03 Delivery & Observability

04 AI-Augmented Engineering

Experience

Development Manager

EMR-Bear, a MedSuite company | Remote | May 2026 - August 2026

Engaged on a bounded 90-day interim assignment to conduct rapid system cartography and operational risk assessment for a multi-tenant healthcare SaaS platform serving 130+ clinics following a founder transition.

Architectural & Risk Cartography: Built a rapid 90-day system inventory and risk model spanning application codepaths, infrastructure dependencies, access boundaries, and operational risks.

Production & Compliance Governance: Protected production stability across 130+ clinical environments by establishing release verification gates across the deployment pipeline,

blocking unvalidated changes prior to HIPAA compliance review and business sign-off.
Engineering Continuity: Created transparent component ownership, visible operational priorities, and collaborative decision rhythms for a distributed U.S.-Mexico engineering team.

Associate Director, Staff Engineer

OneMain Financial | Remote | January 2021 - February 2026

Served as Senior Technical IC and Software Architecture Lead for the Acquisition lane of a regulated financial enterprise, combining hands-on system cartography with production stability, modernization, observability, and cross-team technical direction.

Full-Stack Dependency Cartography: Reconstructed end-to-end customer journeys spanning browser interaction surfaces, API gateways, application middleware, and legacy database layers, establishing a unified operational risk model for incident response and change governance.

Incident Response & Technical Escalation: Served as final escalation point for high-severity outages, translating live root-cause investigations into stronger telemetry, explicit component ownership, and reusable system knowledge.

Lateral State Coupling De-risking: Refactored fragile multi-step Rails workflow state machines, eliminating hidden lateral state mutations across asynchronous boundaries and resolving persistent data integrity failures in core acquisition flows.

Platform Enablement & Responsibility Realignment: Realigned Acquisition engineering around explicit business boundaries, then founded and technically led ACQ Enablement to protect platform reliability, remediation, and modernization work from feature pressure.

Observability & Boundary Instrumentation: Founded and led an enterprise OpenTelemetry initiative, instrumenting legacy microservices and third-party API supply-chain boundaries to convert dark system behavior into quantifiable telemetry and reduced MTTD.

AI-Augmented Systems Investigation: Introduced privacy-conscious local AI workflows for legacy code analysis and engineering support, expanding investigative capacity while keeping human review, security constraints, and source evidence explicit.

Senior Backend Developer

SK Holdings, Inc. | Chicago, IL | January 2019 - December 2020

Led backend stability, performance, and modernization for high-traffic Rails products, improving full-text search, campaign delivery pipelines, and data layer reliability while systems remained under production load.

Search Infrastructure Simplification: Replaced legacy external search clusters (Sphinx) with native MySQL full-text search and optimized relational indices, eliminating external service dependencies and reducing platform infrastructure footprint.

High-Volume Messaging Pipelines: Designed and validated asynchronous batch delivery pipelines via SendGrid and Sidekiq, enforcing IP warmup protocols and deliverability safety for large-scale campaign operations.

Data Engine & Analytics: Engineered complex relational analytical queries across multi-million row content catalogs, powering real-time data visualizations and international media publishing.

Zero-Downtime Platform Upgrades: Led Ruby and Rails framework upgrades across multiple core applications, sequencing database migrations and dependency updates to maintain uptime during platform modernization.

Selected Current Work

AI-Augmented Systems Engineer (Phalanx Duel)

Independent | Remote | 2024 - Present

Designing and building a real-time tactical game platform as a current hands-on laboratory for deterministic systems, product architecture, and controlled AI-assisted engineering.

Designed a deterministic engine with replayable action logs and server-authoritative state transitions, making complex outcomes reproducible and independently verifiable.

Built replay validation, adversarial coverage, and CI verification gates so state transitions can be checked across the complete lifecycle.

Uses controlled agent workflows with bounded tasks, acceptance criteria, automated checks, and human-reviewed outputs rather than treating model output as authority.

Core Capabilities & Technologies

Technical Leadership & Cartography

System Cartography & Knowledge Recovery

Cross-Team Architecture & Technical Direction

Lateral State & Supply Chain Dependency Mapping

Evidence-Backed Technical Decision-Making

Legacy Systems & Modernization

Strangler-Fig Decoupling & API Contracts
Zero-Downtime Data Migration & Integrity
Incident Response & Root Cause Analysis
Distributed Systems & Asynchronous Processing

Delivery & Observability

OpenTelemetry & Observability Architecture
Network Boundary & Access Control Auditing
Platform Enablement & Developer Velocity
Database Performance & SQL Analysis

AI-Augmented Engineering

AI Agent & Automation Interaction Mapping
Codebase Discovery & Domain Rule Extraction
LLM Orchestration & Deterministic Evaluation
Bounded, Human-Reviewed Agent Workflows

Technologies

Ruby on Rails
PostgreSQL (including pgvector)
Sidekiq & Redis
AWS & Cloud Infrastructure
Kubernetes, Docker & Linux

Selected Earlier Experience

Senior Software Developer, BenchPrep | March 2017 - February 2018 | Chicago, IL :
Owned enterprise assessment workflows, leading correctness and platform security in a high-concurrency environment.

Senior Software Developer, ReachLocal | March 2015 - November 2016 | Remote :
Owned API design and modernization strategy, leading incremental legacy migration for a high-volume digital marketing platform.

Open-Source Transition Lead, Coderwall | January 2014 - December 2014 | Remote :
Hired as a contractor by founder Matt Deiters to lead the open-source transition of the Coderwall developer reputation platform, a Y Combinator-backed professional network for software engineers (856 GitHub stars, 304 forks). Delivered security hardening,

proprietary service extraction, infrastructure modernization, and community leadership as the top contributor to the open-source codebase.

Engineering Learning & Development Business Partner, Groupon | July 2011 - May 2013 | Chicago, IL : Owned core backend systems for fraud and taxonomy, leading global engineering enablement during hyper-growth.

Career Foundations

1999 - 2011

Built and operated enterprise integrations, transactional systems, and early web products across consulting and product organizations, establishing the production discipline and community-centered craftsmanship that still shape my leadership.

Revenue-critical commerce: Designed real-time inventory, locking, and fulfillment services for a high-volume ticket marketplace later acquired by a major industry operator.

Consulting and software craftsmanship: Delivered systems across client environments while mentoring engineers, adopting Ruby and Rails, and helping build Chicago's craftsmanship community.

Enterprise and operational systems: Built integration and operational software across large organizations, grounded in .NET, SQL, messaging, production support, and direct user needs.

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